

Fixed objective weights; adaptive micro-batch counts

(a) Fix w ; allocate m

Fixed task weights

Objective w

fixed 



Math Code Instr. Science

Micro-batches in the next step
Counts $m = (m_1, \dots, m_M)$



all tasks present

loss $\times w_i/m_i$

$m_{\min} \leq m_i \leq m_{\max}$

fixed objective

$$E[A | m] = \sum_i w_i \mu_i$$

(b) Measure update noise

reuse gradients from this step

top- k micro-batch gradient $g_{i,s}$

preconditioned gradient

$Dg_{i,s}$; D from optimizer (identity for SGD)

task noise $e_i = \text{Var}(Dg_i)$



no extra forward or backward pass

(c) Allocate compute

reuse the completed step

preconditioned noise e_i

micro-batch cost τ_i



GPAS

$$m_i \propto w_i \sqrt{e_i}$$

$\min V(m)$

optimizer-aware allocation

GPAS: cost-aware mode

$$\min J_C(m); \text{ if } C = 0: m_i \propto w_i \sqrt{e_i / \tau_i}$$

Evaluate: teacher loss vs. steps $\cdot$ GPU hours

next step: update m from current noise and cost; keep w fixed